

Sleuthkit Intro

 **Security Warning:** Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

Challenge Description

Download the disk image and use `mmls` on it to find the size of the Linux partition. Connect to the remote checker service to check your answer and get the flag.

Note: If you are using the webshell, download and extract the disk image into `/tmp`, not your home directory.

Additional details will be available after launching your challenge instance.

Tools Used

- `7z` – Extract compressed files
 - `file` – Determine file type
 - `strings` – Extract printable character sequences
 - `exiftool` – Read metadata
 - `fdisk` – View partition tables
 - `mmls` – Display partition layout (part of The Sleuth Kit)
 - `nc` (netcat) – Connect to remote service
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Complete Solution

Step 1: Download and Extract

Download the disk image named `disk.img.gz` from the challenge page.

Extract the compressed disk image:

```
7z x disk.img.gz
```

This decompresses the gzipped file and creates the raw disk image `disk.img`.

Step 2: Basic File Inspection

Verify the file type and understand its structure:

```
file disk.img
```

Expected output:

```
disk.img: DOS/MBR boot sector; partition 1 : ID=0x83, active, start-CHS (0x0,32,33),
end-CHS (0xc,190,50), startsector 2048, 202752 sectors
```

The output confirms this is a **DOS/MBR boot sector** with a Linux partition (ID `0x83`).

Step 3: Search for Visible Text

Disk images often contain strings embedded directly in the data. Extract all readable character sequences and search for the flag:

```
strings disk.img | grep -i "picoCTF"
```

Output: (no results)

No flag appears yet. We need to analyze the partition table.

Step 4: Inspect Metadata

Examine the disk image metadata:

```
exiftool disk.img
```

Expected output:

```
ExifTool Version Number      : 13.50
File Name                    : disk.img
Directory                   : .
File Size                    : 105 MB
File Modification Date/Time   : 2021:09:21 21:34:53+02:00
File Access Date/Time        : 2026:05:12 17:52:22+02:00
File Inode Change Date/Time   : 2026:05:12 17:52:10+02:00
File Permissions              : -rw-r--r--
Error                        : Unknown file type
```

Nothing suspicious here.

Step 5: View Partitions with fdisk

Use `fdisk` to see the partition structure:

```
fdisk -l disk.img
```

Expected output:

```
Disk disk.img: 100 MiB, 104857600 bytes, 204800 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x5d8b75fc
```

Device	Boot	Start	End	Sectors	Size	Id	Type
disk.img1	*	2048	204799	202752	99M	83	Linux

The Linux partition starts at sector **2048** and ends at sector **204799**. The total number of sectors is **202752**.

Step 6: View Partitions with mmls (The Sleuth Kit)

Use `mmls` to display the partition layout in a more readable format:

```
mmls disk.img
```

Expected output:

```
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors
```

	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table (#0)
001:	-----	0000000000	0000002047	0000002048	Unallocated
002:	000:000	0000002048	0000204799	0000202752	Linux (0x83)

Key finding: The Linux partition has a **length of 202752 sectors** (this matches the `fdisk` output).

Step 7: Connect to the Remote Checker Service

When you launch the challenge instance, a netcat endpoint is provided. The port number may change each time you start the instance (e.g., `58018`).

Connect to the service:

```
nc saturn.picoctf.net 58018
```

Output:

```
What is the size of the Linux partition in the given disk image?
Length in sectors:
```

Step 8: Submit the Answer

Enter the partition size (in sectors) that we found:

```
202752
```

Response:

```
Great work!
```

picoCTF{<REDACTED>}

Final Result

picoCTF{<REDACTED>}

Note: The actual flag is redacted here. In the real challenge, the remote service will return the complete flag after providing the correct answer.

Key Concepts

- **Raw Disk Image:** A sector-by-sector copy of a storage device, preserving partitions and filesystems.
- **MBR (Master Boot Record):** The first sector of a disk that contains the partition table and boot code.
- **Partition Table:** A structure that defines how the disk is divided into logical partitions.
- **Linux Partition (0x83):** The partition type identifier for a native Linux filesystem (e.g., ext4).
- **Sectors:** The smallest addressable unit on a disk (typically 512 bytes).
- **mmls:** A tool from The Sleuth Kit that displays the partition layout of a disk image.
- **Netcat (nc):** A networking utility used to connect to remote services and send/receive data.